

Regulatory Relief and Small Bank Mortgage Lending*

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Abstract

In response to concerns that the Dodd-Frank Act of 2010 over restricted the supply of affordable credit, the late 2010s saw a wave of financial deregulation. This paper studies how two contemporaneous deregulatory changes affected mortgage lending: (i) the creation of a new Qualified Mortgage (QM) category for small banks under the Economic Growth, Regulatory Relief, and Consumer Protection Act (EGRRCPA), and (ii) a sharp reduction in Consumer Financial Protection Bureau (CFPB) enforcement activity. Using Home Mortgage Disclosure Act (HMDA) loan-level data from 2014–2019 and a difference-in-differences style regression comparing banks just above and below the \$10 billion asset threshold at which these policies go into effect, this paper identifies which policy had a larger effect on approval rates, loan retention, and prescreening behaviors. I find that the decline in CFPB oversight had larger effects than the introduction of the EGRRCPA QM on approval rates of effected loans and on their application share. Additional evidence from 2022 reforms to the QM definition suggests that CFPB-supervised banks were more attentive to changes in rules enforced by the CFPB, whereas non-supervised banks continued to rely on earlier “rules of thumb.” These results indicate that the supervisory environment, rather than statutory eligibility for QM status, played the dominant role in shaping lender behavior and have implications for the design and implementation of future deregulation.

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1 Introduction

The 2008 financial crisis was caused in large part by the issuance of risky, subprime mortgages. This sparked a wave of government regulation and legal action. Notably, the Dodd-Frank Act of 2010 introduced regulations aimed at preventing future crises and improving consumer financial protection. However, in the post-crisis period, concerns grew that these regulations imposed excessive compliance costs, especially for small banks, and over-restricted the supply of affordable credit. In response, the late 2010s saw a wave of deregulation and a more relaxed enforcement environment. Mortgages play an essential role in facilitating homeownership, a key means of building wealth in the United States. As policymakers work to balance financial stability with affordable credit access, understanding the effect of these regulatory changes on the supply of mortgage credit is especially important.

The Dodd-Frank Act established the CFPB and tasked it with supervising financial institutions’ consumer lending activity, developing rules to implement federal consumer financial laws, and enforcing consumer protection laws. In 2014, the CFPB began implementing the Ability to Repay (ATR) mortgage rule. The ATR rule requires lenders to make a “reasonable, good faith” determination that the borrower is able to repay the mortgage before issuing credit.¹ As part of the implementation, the CFPB established a class of mortgage with certain lower risk features called Qualified Mortgages (QM) that are automatically presumed to satisfy ATR. This policy led to a substantial contraction of supply in the non-QM market (Defusco et al., 2020). With the goal of improving the availability of mortgage credit, the Economic Growth, Regulatory Relief, and Consumer Protection Act (EGRRCPA) of 2018 created a new type of QM for loans originated and serviced by depository institutions with total assets below \$10 billion. Intuitively, a small mortgage lender that retains a loan on its portfolio is already sufficiently incentivized to determine that the borrower is able to repay the mortgage. At the same time the EGRRCPA went into effect, leadership changes at the CFPB led to a steep drop in the number of enforcement actions. The CFPB serves as the primary consumer compliance regulator for banks with assets above \$10 billion. This article assesses which of these concurrent policy changes—the EGRRCPA QM reform for small banks or broader supervisory easing for larger banks—had a stronger effect on mortgage lending.

¹<https://www.consumerfinance.gov/rules-policy/final-rules/ability-to-pay-qualified-mortgage-rule/>

Using Home Mortgage Disclosure Act (HMDA) loan application data, this paper employs difference-in-differences and triple-difference style regressions comparing banks around the \$10 billion total asset threshold.² The analysis focuses on the market segment most likely to be affected by the EGRRCPA QM: loans that would otherwise not be QMs. Three outcomes are considered: (i) loan approval rates, as a measure of credit access; (ii) portfolio retention rates, which would increase if the EGRRCPA QM made banks more likely to retain otherwise non-QM loans; and (iii) application volumes and shares, which capture changes in pre-screening behavior. Results indicate that the relaxation of CFPB oversight had a larger effect than the creation of the EGRRCPA QM on non-QM lending, specifically loan approval rates and share of non-QM applications received.

It is notable that the EGRRCPA QM, a reform aimed at small banks’ non-QM lending, has a weaker effect than the contemporaneous decline in CFPB oversight, a change not directed at the non-QM market. A plausible mechanism is that CFPB-supervised banks were more attentive to changes in ATR enforcement, while non-supervised banks continued to follow earlier “rules of thumb” from the initial QM definition. Consistent with this, I find that even after the definition of a QM changed for all banks in 2022, the banks that are not supervised by the CFPB continued to adhere more to the old QM definition. These differential responses to deregulation has important implications for the design and implementation of future deregulation.

This project contributes to the literature on the effects of bank regulation on mortgage lending (for recent examples, see [Begley and Srinivasan, 2022](#); [Benzarti, 2024](#); [Buchak et al., 2018](#); [Defusco et al., 2020](#); [Frame et al., 2024](#); [Fuster et al., 2023](#); [Haughwout et al., 2024](#)). Most relevant to this paper, [Defusco et al. \(2020\)](#) finds an increase in the price and a decrease in the supply of credit in the non-QM market following the initial implementation of the ATR rule. [Fuster et al. \(2023\)](#) studies the effect of CFPB oversight on lending behavior, finding a reduction in lending in the Federal Housing Administration (FHA) market, which contains riskier borrowers. By studying how the introduction of the EGRRCPA QM and the relaxation of CFPB oversight affected lender behavior in the non-QM market, this paper advances the understanding of how lenders respond to the ATR rule. There is also a growing literature on the impact of the EGRRCPA ([Anani et al.,](#)

²Throughout this paper, the terms bank and depository institution are used interchangeably, even though the sample contains some credit unions. Controlling for credit union status of the lenders in this study had no impact on the results.

2024; Chronopoulos et al., 2023; Erkens and Gan, 2022; Le and Santos, 2023; Powell, 2022) which this study contributes to by being the first to assess the effects of the EGRRCPA QM on the mortgage lending of small banks using loan-level data.

The remainder of the paper proceeds as follows. Section 2 provides background on the CFPB, Qualified Mortgages, and the EGRRCPA. Section 3 describes the data. Section 4 discusses the empirical approach and section 5 reports the results. Section 6 interprets the results and provides suggestive evidence for a potential mechanism. Section 7 concludes.

2 Institutional Background

2.1 Consumer Lending Regulation

2.1.1 The CFPB

As part of the Dodd-Frank Act, the CFPB was created to serve as a government agency dedicated to the financial protection of consumers. The CFPB has authority over both bank and non-bank lenders financial institutions. The CFPB makes rules to implement federal consumer protection laws that apply to all financial institutions and can pursue enforcement actions against firms that violate these laws. Additionally, the CFPB supervises and examines both non-bank lenders and banks with assets above \$10 billion.

2.1.2 Qualified Mortgages

One way the Dodd-Frank Act sought to ensure financial stability and protect consumers was by requiring that lenders verify a borrower’s ability to make all scheduled payments before originating a mortgage. The CFPB implemented this policy through the Ability to Repay/Qualified Mortgage (ATR/QM) rule, which requires that lenders “make a reasonable, good faith determination of a consumer’s ability to repay a residential mortgage loan according to its terms.”³ The ATR rule mandates that lenders verify and consider certain factors in their underwriting process using reasonably reliable records.⁴ QMs were concurrently introduced as a more straightforward means of

³<https://www.consumerfinance.gov/rules-policy/final-rules/ability-to-pay-qualified-mortgage-rule/>.

⁴Those factors are: “(1) Current or reasonably expected income or assets; (2) current employment status; (3) the monthly payment on the covered transaction; (4) the monthly payment on any simultaneous loan; (5) the monthly payment for mortgage-related obligations; (6) current debt obligations, alimony, and child support; (7) the monthly

satisfying the ATR rule. In addition to following the ATR underwriting factors, QMs have certain well-defined features that make them less risky. When the ATR/QM rule was first implemented in January of 2014, the two main ways for a loan to be a QM were either (1) having a Debt-to-Income (DTI) ratio below 43 percent or (2) being a conforming loan eligible for purchase by the Government Sponsored Enterprises (GSE), Fannie Mae and Freddie Mac. Functionally, conforming loans are loans for amounts below the conforming loan limit.⁵ Throughout this paper, I will refer loans that satisfy at least one of these criteria as “generally QM.” A QM is automatically presumed to comply with the ATR rule, protecting the lender from legal liability.⁶ Appendix A provides more details about the QM classifications and their evolution over time.

Lenders found in violation of the ATR rule can face legal penalties. Within three years of a mortgage’s origination, a borrower can sue the lender for violation of the ATR rule, and if facing foreclosure, the borrower can always sue for violation of the ATR rule. The CFPB estimated average damages of approximately \$30,000 for claims filed within three years, and roughly \$50,000 for cases initiated in response to foreclosure proceedings.⁷ Liability extends beyond the originating lender to secondary purchasers, including holders of mortgage-backed securities. Additionally, the CFPB can pursue enforcement action against lenders engaged in systematic or egregious violations.⁸ Note that while all lenders are subject the ATR/QM rule, banks with assets below \$10 billion are not supervised by the CFPB and are therefore not subject to CFPB legal enforcement actions of this rule. While non-QMs can still satisfy the ATR rule, the opaque phrasing of the rule is a concern for lenders, who must bear the risk of potential legal action. Defusco et al. (2020) find that the initial implementation of the ATR/QM rule in 2014 led to a 10-15 basis point increase in interest rates for non-QM loans and eliminated roughly 15 percent of the non-QM market. Mortgage origination industry reports suggest that underwriting non-QM loans consumes two to three times

debt-to-income ratio or residual income; and (8) credit history.” For the full text of rule, see [Federal Register, Vol. 78, No. 20, pp. 6407–6510](#).

⁵The conforming loan limit is set annually by the Federal Housing Finance Agency (FHFA). For context, in 2025 the conforming loan limit for a single-family home is \$806,500.

⁶The strength of that legal protection varies. For high-priced QMs (with APR at least 1.5 percentage points above the Average Prime Offer Rate (APOR)) the lender has a “rebuttable presumption” of compliance. For non-high-priced QMs, lenders are given a legal “safe harbor,” meaning any law suit claiming ATR violation would fail. All EGRRCPA QMs provide lenders with a legal safe harbor, regardless of price.

⁷See [Federal Register, Vol. 78, No. 20, pp. 6407–6510](#).

⁸For example, in January 2025, the [CFPB sued Vanderbilt Mortgage & Finance](#) for violating the ATR rule. However, the second Trump administration pushed the CFPB to [drop the lawsuit in February](#).

the amount of effort as underwriting similar QM loans (Adelson, 2021). This evidence demonstrates the ATR/QM rule has meaningfully shaped mortgage lending practices.

2.2 Deregulation

2.2.1 EGRRCPA

Enacted in May 2018, the EGRRCPA created a new category of QM for loans that were originated and held on the balance sheet of depository institutions with total assets below \$10 billion. Intuitively, banks are incentivized to ensure the repayment ability of borrowers when they retain the associated credit risk. Small banks especially may be able to discern borrower creditworthiness using soft information and may find regulatory compliance disproportionately burdensome. While these institutions can still originate other types of QMs, they alone are eligible to issue EGRRCPA QMs. This policy would therefore be expected to (weakly) increase the otherwise non-QM lending of banks with assets below \$10 billion. It is possible that treated banks did not adjust their lending in response to the creation of the EGRRCPA QM—the pre-existing QM policy may not have been binding for these banks or these banks might not have been aware of the EGRRCPA QM’s creation. This paper studies how the EGRRCPA’s introduction of a this new QM category affected small bank mortgage lending activity.

Other EGRRCPA Provisions: The EGRRCPA rolled back several post-financial crisis regulations introduced under the Dodd-Frank Act. While the introduction of the EGRRCPA QM is the only provision that relates directly to mortgage lending, several other provisions go into effect at the \$10 billion asset threshold. The primary analysis of this paper, discussed in more detail in section 4, compares the mortgage lending activity of banks with assets above and below the \$10 billion threshold. The discussion below addresses these potentially confounding policies.

1. **Volcker Rule Exemption.** Section 203 of the EGRRCPA exempts from the Volcker Rule banks with assets below \$10 billion that have trading assets and liabilities composing less than 5 percent of their total assets. The Volcker Rule limits banks engagement with proprietary trading and certain hedge funds and private equity funds. This policy is unrelated to bank mortgage lending activity.

2. **Simplified Capital Requirements.** The EGRRCPA exempted banks with assets below \$10 billion that satisfy a simplified leverage ratio⁹ from the more complex Basel III capital requirements. [Srivastava and Vallascas \(2022\)](#) study a similar reform from 2015 and find that it had no effect on banks' overall mortgage lending. Furthermore, because the Basel III requirements risk-weight mortgages by their Loan-to-Value (LTV), this reform is unlikely to affect lending in the generally non-QM markets, which are defined as non-conforming loans with high DTI ratios.¹⁰ To address any broader changes in mortgage lending behavior that vary by bank size, I employ a triple difference strategy that uses generally QM loans as an additional control group to assess how these policy changes affected lending in the generally non-QM market. This strategy is discussed in more detail in section 4 below.

3. **Adjustment of Bank-run Stress Testing Threshold.** The EGRRCPA raised the asset threshold at which bank-run stress tests are required from \$10 billion to \$250 billion. The EGRRCPA also raised the threshold for financial institutions to be designated as systemically important from \$50 billion to \$250 billion in total assets, reducing supervision and mandatory government run stress tests for these banks. Banks with assets between \$10 and \$50 billion did not have to pass government run stress tests prior to EGRRCPA. Rather, they had to publish results from bank-run stress tests on a government website. [Haughwout et al. \(2024\)](#) find that banks did not alter the share of their approved loans that are non-conforming in response to these bank-run stress tests. [Fuster et al. \(2023\)](#) also find that these stress tests had no impact on the composition of banks' mortgage lending. To further ensure that these stress tests are not affecting the results, I compare non-bank lenders, who were never subject to stress tests, to banks with assets below \$10 billion as a robustness check.

2.2.2 CFPB Enforcement Changes

The CFPB's approach to supervision and enforcement actions altered substantially under new leadership appointed during the first Trump administration. In late November of 2017, Trump appointed a new Acting Director of the CFPB, Mick Mulavaney, who was followed by a new permanent

⁹Leverage ratio refers to the ratio of equity capital to assets.

¹⁰A detailed breakdown of Basel III mortgage requirements is available at <https://www.butlersnow.com/news-and-events/basel-iii-impact-residential-mortgages>.

Director, Kathy Kraninger, in December of 2018. Under this new leadership, CFPB enforcement actions declined by 70 percent and associated restitution payments to consumers declined by 85 percent overall—99 percent for mortgages (Peterson, 2019). The bureau also reversed a pattern of consistent employment growth with a hiring freeze and a 15 percent staff reduction. In congressional testimony from April 2018, Acting Director Mulvaney spoke publicly about this change in enforcement, saying that the CFPB’s “new strategic priorities are to recognize free markets and consumer choice.” Since \$10 billion is the asset threshold above which banks are supervised by the CFPB and the ATR/QM rule is implemented and enforced by the CFPB, larger banks may increase non-QM lending in response to the relaxed enforcement environment.

Since the EGRRCPA QM went into effect in May 2018 and the CFPB leadership changed in November 2017 with the new Director formally announcing the bureau’s intent to scale back enforcement and supervisory efforts in April 2018, the timing of the policies is extremely similar and indistinguishable in annual data. The EGRRCPA QM was a deregulation designed to increase the generally non-QM lending of small banks. The reduction of CFPB enforcement actions for larger banks was not specifically aimed at non-QM lending, but it may also have led to an increase in the non-QM lending of larger banks since the CFPB implements and enforces the ATR/QM rule and directly supervises banks with assets above \$10 billion. Since both policies could cause banks to increase their non-QM mortgage lending, this paper assesses which policy had a larger effect on non-QM mortgage lending by comparing banks above and below the \$10 billion asset threshold.

3 Data

3.1 Sources and Sample Construction

My primary data source is the publicly available Home Mortgage Disclosure Act (HMDA) Loan/Application Register (LAR) from 2014 to 2019. I begin the sample in 2014 since that is when the ATR/QM rule went into effect, and I stop in 2019 to avoid confounding factors from the COVID-19 pandemic as well as subsequent changes to the QM categories. The HMDA data provide comprehensive information on mortgage applications, including lender identity, lending decision, loan

characteristics, and borrower characteristics for nearly all mortgage lenders in the United States.¹¹ I use the HMDA Lender Panel for information on lender total assets.¹²

To evaluate the effects of the EGRRCPA QM policy implemented at the \$10 billion in total assets threshold, I restrict the sample to depository institutions with assets between \$6 and \$20 billion in 2017. I set the lower bound to be greater than \$5 billion because of other provisions of the EGRRCPA that affect banks with assets below \$5 billion. I chose an upper bound below \$50 billion because of changes in the government run stress test requirements that applied to banks with assets above \$50 billion. I experiment with several different asset ranges for the analysis and find that the results are not sensitive to the chosen values. Appendix figure B1 plots the distribution of bank assets in 2017 around the \$10 billion threshold. I use the total assets in 2017 to address potential endogeneity concerns caused by the fact that banks may adjust their balance sheet size in response to the policy changes.

I exclude loan applications that were incomplete or withdrawn, and I further restrict the sample to conventional mortgages, meaning I discard applications for programs run by the Federal Housing Administration (FHA), Department of Veterans Affairs (VA), Farm Service Agency, or Rural Housing Service.¹³ Lastly, I limit the analysis to first-lien home purchase and refinancing mortgages. This leaves 104 banks (70 with assets below \$10 billion and 34 with assets above) receiving 2,322,443 loan applications (1,072,780 of which were to banks with assets below \$10 billion while the remaining 1,249,663 were to banks with assets above). Through the remainder of this paper, I refer to this as the “full sample.”

Because banks with assets below \$10 billion are still able to issue generally QM loans, we would expect the policy to affect their lending behavior for loans that would otherwise not be QMs. That is, we are interested in seeing how this policy affects lending of non-conforming loans with DTI above 43 percent. Beginning in 2018, information on the DTI of a loan and whether it is conforming

¹¹While the reporting thresholds change somewhat over time, in 2018 depository institutions report to HMDA if they have at least one branch in a metropolitan statistical area (MSA), originated at least 25 closed-end mortgage loans in the preceding two calendar years, and have assets above \$45 million. The HMDA data are estimated to cover over 95 percent of all non-rural first-lien mortgages (Avery et al., 2017).

¹²To ensure consistent identification of lenders across years, I utilize Zach Babat’s `hmda_harmonizer` repository available at https://github.com/ZachBabat/hmda_harmonizer/.

¹³Most of these non-conventional loans are considered QMs because of the agency issuing them, see appendix A for details.

(i.e., below the conforming loan limit) is reported. Unfortunately, this information is not included for earlier years of the data.

To identify conforming loans in the pre-2018 data, I merge in data from the Federal Housing Finance Agency (FHFA) on the conforming loan limit in each county and year. I use the one-unit dwelling conforming loan limit to construct an indicator for whether a loan is conforming. Information on the number of units in a dwelling is not available in the HMDA data before 2018, but 97.53 percent of the full sample in 2018 and 2019 were applications for 1 unit dwellings. Thus, while suffering from some measurement error, this approximation is fairly straightforward.

Identifying high DTI loans is a more challenging task. The DTI ratio is calculated as the applicant’s total monthly debt to total monthly income. While total annual income is included in the HMDA data, total debt is not. However, the amount of the mortgage loan is available in the data. I use this as a proxy for total debt and calculate the Loan-to-Income (LTI) ratio. 75.2 percent of applications in the full sample for 2018 and 2019 with non-missing DTI information have a DTI below 43 percent. Using the entire HMDA LAR dataset for 2014 to 2019, I identify loans in the top quartile for LTI values. 77.37 percent of loans in the full sample for all years are in the bottom three quartiles for LTI values. Thus, using the highest quartile of LTI for all applications seems to be a reasonable proxy for DTI above 43 percent.¹⁴ To study the effect of the creation of EGRPCAs on bank lending behavior, I focus much of my analysis on the subsample of loans above the conforming loan limit with LTI ratios in the top quartile of all applications. I refer to this as the “general non-QM sample,” and it is 2.67 percent of the full sample.

Because the deregulations studied in this paper are applied to banks, portions of the analysis directly consider bank level behavior. The full sample of loan level data are aggregated to the bank-year level to create the “bank-year level sample.”

¹⁴To assess how justifiable it is to use high LTI & non-conforming loan applications as a stand in for DTI above 43 percent & non-conforming loan applications, I construct an indicator for DTI above 43 percent & non-conforming in the sample for 2018 and 2019. 74.4 percent of DTI above 43 percent & non-conforming applications are classified as high LTI & non-conforming, while 1.68 percent of DTI below 43 percent or conforming applications are classified as high LTI & non-conforming. Clearly, the high LTI & non-conforming loan variable is an imperfect proxy for generally non-QM loans over this period.

3.2 Descriptive Statistics

Descriptive statistics for the full sample and general non-QM sample are reported in table 1. Loan approval rates for banks with assets above and below \$10 billion are similar for the full sample, between 82 and 85 percent. However, for the general non-QM sample, larger banks have much lower approval rates on average than the smaller banks, 71 percent compared to 82. Smaller banks are also more likely to retail loans on their portfolio. For the full sample, the share of approved loans that are retained on the banks balance sheet is 54 percent for banks with assets \$6 to \$10 billion whereas it is only 39 percent for banks with assets \$10 to \$20 billion. This gap narrows for high LTI & non-conforming loans to 87 percent for small banks and 83 percent for large banks, likely due to the fact that non-QM loans are harder to securitize. While most of the other variables are similar across bank size, small banks receive a higher share of applications for manufactured homes, and manufactured home applications are less likely to be generally non-QM. Lastly, it is worth noting that the vast majority of the market (93 percent of the full sample) is below the conforming loan limit, providing important context for the limited scope of this study and impact of this policy.

Table 2 provides additional descriptive statistics for select variables in the bank-year level sample. While banks with assets above \$10 billion receive a larger number of generally non-QM applications, these generally non-QM applications make up a greater share of the total applications received by banks with assets below \$10 billion. Averages for variables such as loan approval rates, retention rates, and characteristics are not reported for the bank-year level sample because they align very closely with the averages reported for the loan level data in table 1.

Table 1: Descriptive Statistics by Bank Size, Loan Level

	Full Sample		General Non-QM Sample	
	Mean	SD	Mean	SD
Assets_{b,2017} ≥ \$10b				
Loan Approved	0.827	0.378	0.709	0.454
Loan Retained*	0.388	0.487	0.832	0.374
Joint Application	0.501	0.500	0.590	0.492
Conforming Loan	0.937	0.243	—	—
Refinancing	0.490	0.500	0.459	0.498
Manufactured Home	0.00830	0.0907	0.000702	0.0265
Loan Amount (000s)	259.1	244.9	861.7	822.3
Applicant Income (000s)	129.4	1009.3	187.1	109.7
Black	0.0358	0.186	0.0160	0.125
Hispanic	0.0887	0.284	0.0556	0.229
<i>Observations</i>	<i>1,249,663</i>		<i>29,895</i>	
Assets_{b,2017} < \$10b				
Loan Approved	0.848	0.359	0.815	0.388
Loan Retained*	0.543	0.498	0.867	0.340
Joint Application	0.511	0.500	0.523	0.499
Conforming Loan	0.919	0.273	—	—
Refinancing	0.458	0.498	0.391	0.488
Manufactured Home	0.0315	0.175	0.000654	0.0256
Loan Amount (000s)	255.9	344.5	991.2	1315.8
Applicant Income (000s)	130.8	665.9	211.2	177.4
Black	0.0354	0.185	0.0159	0.125
Hispanic	0.0943	0.292	0.0517	0.221
<i>Observations</i>	<i>1,072,780</i>		<i>32,088</i>	
All				
Loan Approved	0.837	0.370	0.764	0.425
Loan Retained*	0.461	0.498	0.851	0.356
Joint Application	0.506	0.500	0.556	0.497
Conforming Loan	0.929	0.257	—	—
Refinancing	0.475	0.499	0.423	0.494
Manufactured Home	0.0190	0.137	0.000678	0.0260
Loan Amount (000s)	257.6	295.1	928.7	1107.5
Applicant Income (000s)	130.1	867.7	199.6	149.1
Black	0.0356	0.185	0.0159	0.125
Hispanic	0.0913	0.288	0.0535	0.225
<i>Observations</i>	<i>2,322,443</i>		<i>61,983</i>	

Notes: *Loan retained refers to loans that are not sold by banks in the calendar year. This variable is only available for approved loans. Accordingly, in the full sample, there are 1,033,280 observations at banks with $Assets_{b,2017} \geq \$10b$ and 910,121 observations at banks with $Assets_{b,2017} < \$10b$, and in the general non-QM sample there are 21,181 observations at banks with $Assets_{b,2017} \geq \$10b$ and 26,148 observations at banks with $Assets_{b,2017} < \$10b$. The full sample refers to all conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications from 2014 to 2019 at banks with assets between \$6 and \$20 billion in 2017. General Non-QM sample refers to loans that are non-conforming (above the conforming loan limit) and have LTI in the top quartile among all HMDA/LAR records over the sample period.

Table 2: Descriptive Statistics by Bank Size, Bank-Year Level

	Mean	SD
Assets_{b,2017} ≥ \$10b		
Number of General Non-QM Applications	151.8	262.8
General Non-QM Application Share	0.0400	0.0453
<i>Observations</i>	<i>197</i>	
Assets_{b,2017} < \$10b		
Number of General Non-QM Applications	77.69	150.8
General Non-QM Application Share	0.0465	0.0844
<i>Observations</i>	<i>413</i>	
All		
Number of General Non-QM Applications	101.6	197.0
General Non-QM Application Share	0.0444	0.0741
<i>Observations</i>	<i>610</i>	

Note: Based on bank-year level values aggregated from the full sample of loan-level data for all conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications from 2014 to 2019 at banks with assets between \$6 and \$20 billion in 2017. General Non-QM refers to loans that are non-conforming (meaning above the conforming loan limit) and have LTI in the top quartile among all HMDA/LAR records over the sample period.

4 Empirical Strategy

4.1 Loan level regressions

I utilize a difference-in-differences style regression to study how depository institutions altered their mortgage lending in response to the creation of the EGRRCPA QM and the reduction of CFPB enforcement activity. On the sample of general non-QM loans, I estimate the equation:

$$Approved_{i,b,t} = \alpha + \beta \mathbb{1}\{Assets_{b,2017} < \$10b\} \times Post_t + \gamma X_i + \delta_t + \theta_b + \epsilon_{i,b,t} \quad (1)$$

where, for loan application i to bank b in year t , the outcome considered is whether the loan is approved. $\mathbb{1}\{Assets_{b,2017} < \$10b\}$ indicates the application was made at a bank with assets below \$10 billion in 2017, $Post_t$ is an indicator equal to one if the application is filed in 2018 or later, X_i is a vector of loan, property, and applicant characteristics, δ_t are year fixed effects, and θ_b are bank fixed effects. Since treatment is assigned at the bank level, where loan outcomes are likely correlated, I cluster the standard errors by bank.

While this regression is similar to a difference-in-differences regression, it is not the standard set-up with a treatment and control group. Rather, both groups—banks above and below \$10 billion

in assets—are treated by different policies. The small banks are treated by the creation of the EGRRCPA QM while the large banks are treated by the weakening of their consumer protection regulator, the CFPB. Thus, although the equation looks like a traditional difference-in-differences regression with small banks as the “treated” group, it does not allow for the identification of the magnitude of each policy’s effect on approval rates. Instead, we are able to determine which policy had a larger effect on the outcome.

The coefficient of interest is β , which measures the change in approval rates for otherwise non-QM loans at small banks (assets \$6 to \$10 billion) relative to the change at larger banks (assets \$10 to \$20 billion) following the creation of the EGRRCPA QM and the reduction in CFPB enforcement activity. Holding loan characteristics, time-invariant differences across banks, and aggregate changes in approval rates over time constant, the sign of β reveals which policy had a larger causal effect on the generally non-QM lending of banks. Since we would expect both policies to (weakly) increase mortgage lending by reducing the regulations faced by banks, a value of β greater than zero suggests that the creation of the EGRRCPA QM had a larger effect on approval rates for generally non-QM loans than the reduction of CFPB enforcement. A value of β less than zero suggests that the reduction in enforcement activity by the CFPB had a larger effect than the creation of the EGRRCPA QM. To interpret these estimates as the causal net effect of the policies, the identifying assumptions are that (a) banks above and below the \$10 billion asset threshold would have evolved along parallel trends absent the policies and (b) both group of banks would have responded to the treatment that they were not assigned in the same way as the group that was assigned that treatment.

In addition to affecting the approval of a loan application, the policies may also affect whether or not banks choose to retain the loan on their portfolio after it has been originated. The EGRRCPA QM specifically requires that the loan be kept on the balance sheet of the originating bank to maintain its QM status. Thus, we would expect banks using their new QM to weakly increase the share of otherwise non-QM loans that are retained on their portfolio. However, since generally non-QM loans are already more likely to be retained by banks (table 1) because they are not eligible for purchase by the GSEs, it is also possible this policy had no effect on banks decisions to retain generally non-QM loans. It is not clear how the reduction in CFPB supervision would have any

impact on banks loan retention decisions. The CFPB supervises banks consumer lending, not their portfolio behavior. Therefore, under the assumption that the CFPB enforcement relaxation did not affect banks' loan retention decisions, I estimate the following difference-in-differences specification on approved loans in the general non-QM sample:

$$Retained_{i,b,t} = \alpha + \beta \mathbb{1}\{Assets_{b,2017} < \$10b\} \times Post_t + \gamma X_i + \delta_t + \theta_b + \epsilon_{i,b,t} \quad (2)$$

where, for loan application i to bank b in year t , the outcome considered is whether the loan is retained on the balance sheet of the issuing bank. As in equation 1, $\mathbb{1}\{Assets_{b,2017} < \$10b\}$ indicates the application was made at a bank with assets below \$10 billion in 2017, $Post_t$ is an indicator equal to one if the application is filed in 2018 or later, X_i is a vector of loan, property, and applicant characteristics, δ_t are year fixed effects, and θ_b are bank fixed effects. Standard errors are clustered by bank. The coefficient β captures the effect of the EGRRCPA QM creation on the share of otherwise non-QM loans retained by treated banks, relying on the identifying assumption of parallel trends.

There may be concerns that these difference-in-differences style regressions conflate the effect of the policies on generally non-QM loans with broader changes in lending over the same period. These concerns are especially salient given the simplified capital requirements and adjustments to stress testing thresholds also included in the EGRRCPA (discussed in section 2.2.1) and given the broader effects of the CFPB regulatory relaxation on mortgage lending behavior. To address this, I estimate a triple difference style regression that utilizes generally QM loans as an additional control group.¹⁵ This controls for any contemporaneous shocks to bank mortgage lending behavior that vary by bank size, allowing us to isolate the effect of the policies on generally non-QM lending. Using the full sample, I estimate the regression:

$$\begin{aligned} y_{i,b,t} = & \alpha + \beta_1 \mathbb{1}\{Assets_{b,2017} < \$10b\} \times Post_t \times GeneralNonQM_{i,b,t} + \\ & \beta_2 Post_t \times \mathbb{1}\{Assets_{b,2017} < \$10b\} + \beta_3 Post_t \times GeneralNonQM_{i,b,t} + \\ & \beta_4 \mathbb{1}\{Assets_{b,2017} < \$10b\} \times GeneralNonQM_{i,b,t} + \beta_5 GeneralNonQM_{i,b,t} + \gamma X_i + \delta_t + \theta_b + \epsilon_{i,b,t} \end{aligned} \quad (3)$$

¹⁵Generally QM loans refers to loans that are conforming or have a DTI below 43 percent. This paper proxies for DTI with LTI. Thus, generally QM loans are loans below the conforming loan limit with LTI below the 75th percentile for the HMDA LAR data during the sample period 2014 - 2019.

where, for loan application i to bank b in year t , $GeneralNonQM_{i,b,t}$ is an indicator for the loan being generally non-QM. The outcomes $y_{i,b,t}$ considered are application approval, and for approved loans, whether they are retained. Standard errors are clustered at the bank level. The coefficient of interest is β_1 , which captures the differential change in the outcome at small banks relative large banks for generally non-QM loans relative to generally QM loans following the creation of the EGRRCPA QM and the reduction in CFPB enforcement activity.

4.2 Bank level regressions

While loan approval rates are an important measure of access to credit, the policies studied in this paper are applied at the bank level and therefore may have direct effects on bank behavior. Importantly, loan applications received by banks may measure the pre-screening processes of banks. Pre-screening is potentially a more informative measure of mortgage supply than approval rates. Banks may alter their prescreening process in response to the policy changes, with loan approval rates changing as a secondary effect. We would expect both the EGRRCPA QM and the reduction in CFPB oversight activity to lead to an increase in the amount of generally non-QM loans that are pre-screened by banks, since both of these policies would make banks (weakly) more comfortable with generally non-QM lending.

To assess which policy had a greater impact on the amount of generally non-QM loan applications received, I estimate the following difference-in-differences style regression using the bank-year level sample (aggregated from the loan level data):

$$y_{b,t} = \alpha + \beta \mathbb{1}\{Assets_{b,2017} < \$10b\} \times Post_t + \gamma X_b \times \delta_t + \delta_t + \theta_b + \epsilon_{b,t} \quad (4)$$

where, for bank b in year t , the outcomes considered are $\ln(\text{Number of generally non-QM applications})$ and the share of applications that are generally non-QM. As in the loan level regressions, $\mathbb{1}\{Assets_{b,2017} < \$10b\}$ indicates the bank has assets below \$10 billion in 2017, $Post_t$ equals 1 if the year is 2018 or later, δ_t are year fixed effects, and θ_b are bank fixed effects. X_b is a vector of pre-period (2014 - 2017) bank level averages of covariates. These covariates are interacted with year to control bank trends in the covariates. Standard errors are clustered by bank. As with equation 1, the sign of β reveals which policy had a larger impact on bank pre-screening behavior.

A positive β suggests the EGRRCPA QM creation increased the outcome more than the CFPB regulatory easing whereas a negative β suggests the opposite. To interpret the estimates as the causal net effect of the policies on the outcome, the identifying assumptions are that both groups of banks would have evolved along parallel trends absent the policies and that both group of banks would have responded to the other group’s treatment the same way. Additionally, interpreting these difference in applications as a measure of supply and the result of bank behavior requires the assumption that demand for mortgages at both groups of banks evolved similarly over this time. Given that banks with assets between \$6 and \$20 billion are much smaller than large national banks and even large regional banks, this assumption is not unreasonable.¹⁶

5 Results

5.1 Mortgage Approval

5.1.1 Graphical Evidence

Before proceeding with regression results, I assess the reasonability of the parallel trends assumption using graphical evidence. Figure 1 plots the observed approval rates by bank size for generally non-QM loan applications and for generally QM loan applications in the full sample. While the approval rates for generally QM loans (dashed lines) are nearly identical throughout the sample period, the approval rate levels for generally non-QM loans (solid lines) vary substantially by bank size. This may be due to larger banks in the sample having a business model which relies less on retaining the loans they originate.¹⁷ This difference in levels is not a threat to identification so long as the approval rates follow parallel trends. 2014 is the only year in which the approval rates for generally non-QM loans at the different bank sizes do not appear to be moving in parallel.¹⁸ Thus, as a robustness check, I perform the analysis with 2014 omitted. Additionally, in appendix C, I perform the analysis with propensity score re-weighting to improve comparability between banks and address pre-trends. With this visual analysis in mind, we can proceed into regression results

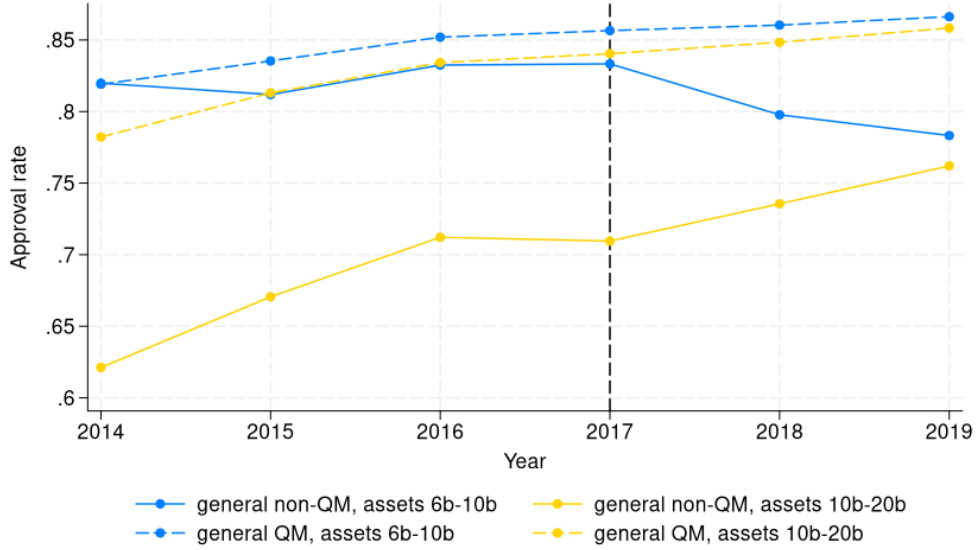
¹⁶For context, Wells Fargo (a large national bank) had \$1.951 trillion in assets in 2017, and Citizens Financial Group (a New England and Mid-Atlantic regional bank) had \$152 billion in assets in 2017.

¹⁷The descriptive statistics in table 1 show that banks with assets above \$10 billion are less likely to retain loans than banks with assets below \$10 billion.

¹⁸An event study style regression for loan approval confirms that 2014 has a statically significant pre-trend even when loan and borrower characteristics are controlled for.

for loan approval rates.

Figure 1: Mortgage Approval Rate Trends by Bank Size and Market Segment



Note: The figure plots the observed means of approval rate by bank size (treatment status) and market segment (general non-QM and general QM) for the full sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications at banks with assets between \$6 and \$20 billion in 2017. General non-QM refers to loans that are above the conforming loan limit and have LTI in the top quartile among all HMDA/LAR records over the sample period.

5.1.2 Regression Results

Table 3 presents regression results for loan approval rates. Columns (1)-(3) report results for the difference-in-differences style regression equation 1 estimated using generally non-QM loan applications. The estimates for β are statistically significant and negative across all specifications. The magnitude of the coefficient shrinks slightly when loan and borrower characteristics are included as covariates (column 2), and the magnitude falls further when 2014 observations are omitted from the analysis, with the significance falling to the 10 percent level (column 3). The persistently negative estimate for the coefficient on $\mathbb{1}\{Assets < \$10b\} \times Post$ suggests that the loosening of the CFPB's regulatory approach had a greater effect on the approval rate of generally non-QM loans than the creation of the EGRRCPA QM.

The results for the triple difference style regression equation 3 estimated using the full sample are reported in columns (4)-(6) of table 3. This specification accounts for broader changes in lending by bank size over the sample period. This is especially important to consider because of the

changing capital requirements for banks with assets below \$10 billion introduced in the EGRRCPA (discussed in section 2.2.1). The estimates of β_1 are statistically significant and negative across all specifications, suggesting that the creation of the EGRRCPA QM had a smaller effect than the reduction in CFPB enforcement activity on the differential change in approval rates for generally non-QM loans relative to generally QM loans.

These results are robust to several tests. Because the EGRRCPA also adjusted a stress testing threshold that affected banks with assets above \$10 billion (discussed in section 2.2.1), I perform the analysis with non-bank lenders in place of banks with assets above \$10 billion and find similar results.¹⁹ To approve comparability between banks of different sizes and further address potential pre-trends, I also propensity score weight the banks. The details for this process are described in appendix C, and table C1 reports the results of the reweighted analysis. The point estimates remain negative and the triple difference style specification remains statistically significant. Taken together, these findings indicate that the relaxation of CFPB enforcement outweighed the introduction of the EGRRCPA QM in shaping approval rates for non-QM loans.

5.2 Loan Retention

5.2.1 Graphical Evidence

I now consider the effect of the EGRRCPA QM creation on loan retention. Figure 2 plots the observed loan retention rates by bank size for generally non-QM loan applications and for generally QM loan applications in the full sample. While smaller banks are more likely than larger banks to retain generally QM loans (dashed lines), the retention rates follow similar trends. The share of generally non-QM loans (solid lines) that are retained are similar levels across banks of different sizes, likely because generally non-QM loans cannot be sold to the GSEs due to their non-conforming status. However, the pre-treatment period trends for generally non-QM loans do not appear to evolve in parallel across bank size. An event study style regression for loan retention confirms that 2014 has a statically significant pre-trend even when loan and borrower characteristics are included as covariates. To address this, I estimate a version of the analysis with 2014 data excluded. I also

¹⁹Non-bank lenders were never subject to stress tests, are not eligible for the EGRRCPA QM, but are supervised by the CFPB. Due to the differences in the business models between banks and non-bank lenders, non-bank lenders are not used in my primary analysis.

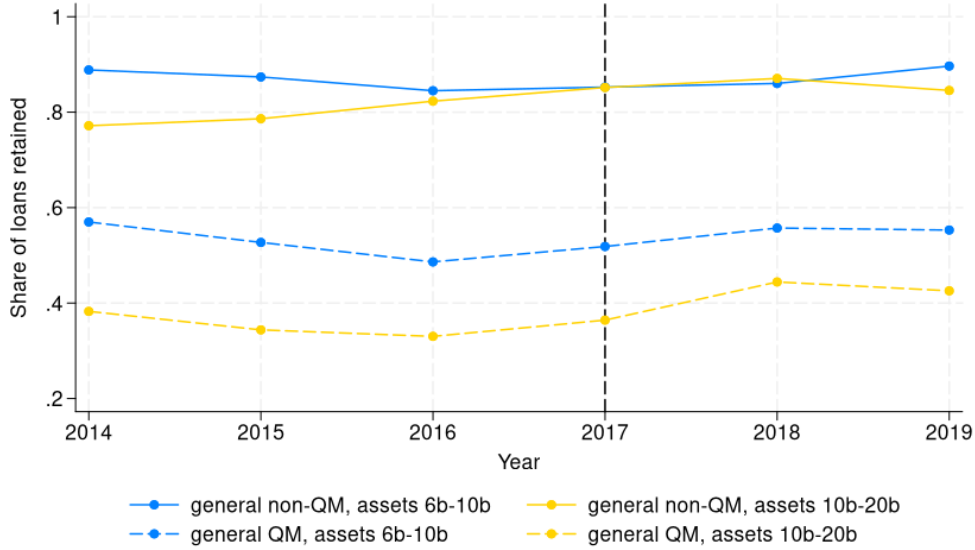
Table 3: Mortgage Approval Rate Regressions

	Diff-in-Diff Style			Triple Diff Style		
	(1)	(2)	(3)	(4)	(5)	(6)
$1 \{Assets < \$10b\} \times Post \times GenNonQM$				-0.074*** (0.027)	-0.069*** (0.024)	-0.061*** (0.021)
$1 \{Assets < \$10b\} \times Post$	-0.047** (0.022)	-0.036** (0.018)	-0.027* (0.016)	-0.000 (0.009)	0.001 (0.008)	0.005 (0.007)
$Post \times GenNonQM$				0.031 (0.021)	0.027 (0.019)	0.023 (0.017)
$1 \{Assets < \$10b\} \times GenNonQM$				0.094*** (0.035)	0.087*** (0.032)	0.083*** (0.031)
$GenNonQM$				-0.139*** (0.033)	-0.137*** (0.023)	-0.134*** (0.022)
Joint Application		0.006 (0.005)	0.005 (0.005)	0.006** (0.003)	0.005** (0.003)	
Refinancing		-0.125*** (0.010)	-0.124*** (0.011)	-0.082*** (0.011)	-0.079*** (0.011)	
Manufactured Home		-0.210** (0.093)	-0.235*** (0.084)	-0.179*** (0.042)	-0.179*** (0.042)	
Black		-0.089*** (0.012)	-0.092*** (0.013)	-0.101*** (0.010)	-0.098*** (0.010)	
Hispanic		-0.036*** (0.008)	-0.040*** (0.008)	-0.050*** (0.008)	-0.049*** (0.008)	
Constant	0.771*** (0.004)	0.823*** (0.006)	0.826*** (0.006)	0.839*** (0.002)	0.887*** (0.008)	0.891*** (0.007)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE		Yes	Yes		Yes	Yes
Income Centile		Yes	Yes		Yes	Yes
Amount Centile		Yes	Yes		Yes	Yes
2014 Omitted			Yes			Yes
Observations	61,981	61,610	54,465	2,322,443	2,322,443	1,999,236
R-squared	0.091	0.228	0.225	0.055	0.122	0.119

Note: Robust standard errors clustered at the bank level in parentheses, *** p<0.01, ** p<0.05, * p<0.1. Columns (1)-(3) report results for equation 1 estimated using general non-QM loan applications in the sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications from 2014 to 2019 at lenders with assets between \$6 and \$20 billion in 2017. General non-QM refers to loans that are non-conforming (above the conforming loan limit) and have LTI in the top quartile among all HMDA/LAR records over the sample period. Columns (4)-(6) report results for equation 3 with loan approval as the outcome, estimated using the full sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications from 2014 to 2019 at lenders with assets between \$6 and \$20 billion in 2017.

propensity score re-weight the banks to improve comparability in appendix C.

Figure 2: Mortgage Retention Rate Trends by Bank Size and Market Segment



Note: The figure plots the observed means of loan retention rate by bank size (treatment status) and market segment (general non-QM and general QM) for the full sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications at banks with assets between \$6 and \$20 billion in 2017. General non-QM refers to loans that are above the conforming loan limit and have LTI in the top quartile among all HMDA/LAR records over the sample period. Loans are considered retained if the bank does not sell them within the calendar year of origination.

5.2.2 Regression Results

Because EGRRCPA QMs must remain on the lender’s balance sheet to maintain their QM status, if treated banks are indeed issuing such loans, we might expect the share of generally non-QM loans that they retain on their balance sheet to increase. However, since generally non-QM loans are already less likely to be sold, it is possible that the policy did not affect banks retention decision for these loans. Because CFPB regulation does not extend to bank’s balance sheet decisions, these regressions are a more straightforward difference-in-differences analysis with small banks being treated by the EGRRCPA QM policy and larger banks serving as the control group. Table 4 reports results from the difference-in-differences equation 2 (columns 1 - 3) and the triple difference equation 3 with loan retention as the outcome (columns 4 - 6). Column (1) reports baseline difference-in-differences results, column (2) includes borrower, loan, and property controls, and column (3) excludes 2014 data from the analysis. Baseline triple difference results are reported in column (4) with borrower, loan, and property characteristics added in column (5) and 2014 excluded in column (6). Across all specifications, the analysis yields no statistically significant evidence of

the EGRRCPA QM affecting banks' decisions to retain generally non-QM loans. Propensity score re-weighted results reported in table C2 are similarly insignificant.

Table 4: Mortgage Retention Rate Regressions

	Diff-in-Diff			Triple Diff		
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}\{Assets < \$10b\} \times Post \times GenNonQM$				-0.009 (0.046)	-0.016 (0.043)	-0.005 (0.045)
$\mathbb{1}\{Assets < \$10b\} \times Post$	-0.008 (0.026)	-0.013 (0.025)	0.002 (0.027)	-0.019 (0.018)	-0.015 (0.017)	-0.014 (0.017)
$Post \times GenNonQM$				-0.007 (0.039)	-0.014 (0.037)	-0.026 (0.037)
$\mathbb{1}\{Assets < \$10b\} \times GenNonQM$				0.014 (0.074)	0.022 (0.062)	0.016 (0.065)
$GenNonQM$				0.340*** (0.068)	0.131*** (0.048)	0.141*** (0.050)
Joint Application		-0.008 (0.005)	-0.009 (0.006)		-0.010*** (0.003)	-0.010*** (0.003)
Refinancing		-0.005 (0.020)	-0.004 (0.022)		0.028** (0.012)	0.028** (0.012)
Manufactured Home		0.033 (0.025)	0.060* (0.036)		0.174*** (0.037)	0.170*** (0.036)
Black		0.011 (0.011)	0.011 (0.012)		0.054*** (0.014)	0.055*** (0.015)
Hispanic		-0.022*** (0.008)	-0.019** (0.008)		0.019** (0.008)	0.020** (0.009)
Constant	0.852*** (0.004)	0.860*** (0.009)	0.857*** (0.011)	0.455*** (0.003)	0.447*** (0.007)	0.444*** (0.007)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	No	Yes	Yes	No	Yes	Yes
Income Centile	No	Yes	Yes	No	Yes	Yes
Amount Centile	No	Yes	Yes	No	Yes	Yes
2014 Omitted	No	No	Yes	No	No	Yes
Observations	47,328	46,996	41,727	1,943,401	1,943,358	1,685,705
R-squared	0.285	0.321	0.322	0.325	0.389	0.379

Note: Robust standard errors clustered at the bank level in parentheses, *** p<0.01, ** p<0.05, * p<0.1. Columns (1)-(3) report results for equation 2 estimated using general non-QM approved loans in the sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications from 2014 to 2019 at lenders with assets between \$6 and \$20 billion in 2017. General non-QM refers to loans that are non-conforming (above the conforming loan limit) and have LTI in the top quartile among all HMDA/LAR records over the sample period. Columns (4)-(6) report results for equation 3 with loan retention as the outcome, estimated using the full sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications from 2014 to 2019 at lenders with assets between \$6 and \$20 billion in 2017. Loans are considered retained if the bank does not sell them within the calendar year of origination.

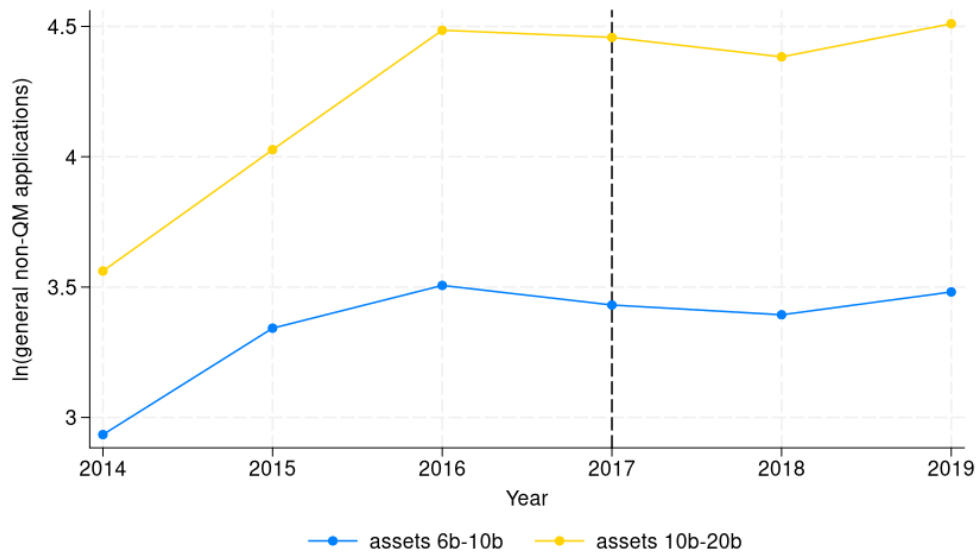
5.3 Bank level

5.3.1 Graphical Evidence

To assess the effects of the policies on banks' pre-screening processes, this section analyses the number and share of generally non-QM loan applications received by banks. Figure 3 plots the

average of the natural log of general non-QM applications received by banks each year over time for banks above and below the \$10 billion asset threshold. Figure 4 plots the average share of applications received by banks each year that are general non-QM applications for banks above and below the \$10 billion asset threshold. While the larger banks receive a higher number of general non-QM applications, these applications make up a higher share of the total applications received by smaller banks in the pre-treatment period. Following treatment, larger banks have a greater share of applications that are general non-QM compared to smaller banks. While the trends evolve fairly similarly, I employ propensity score reweighting to improve the pre-trends in appendix C.

Figure 3: Bank Level $\ln(\text{General Non-QM Applications})$ Trends

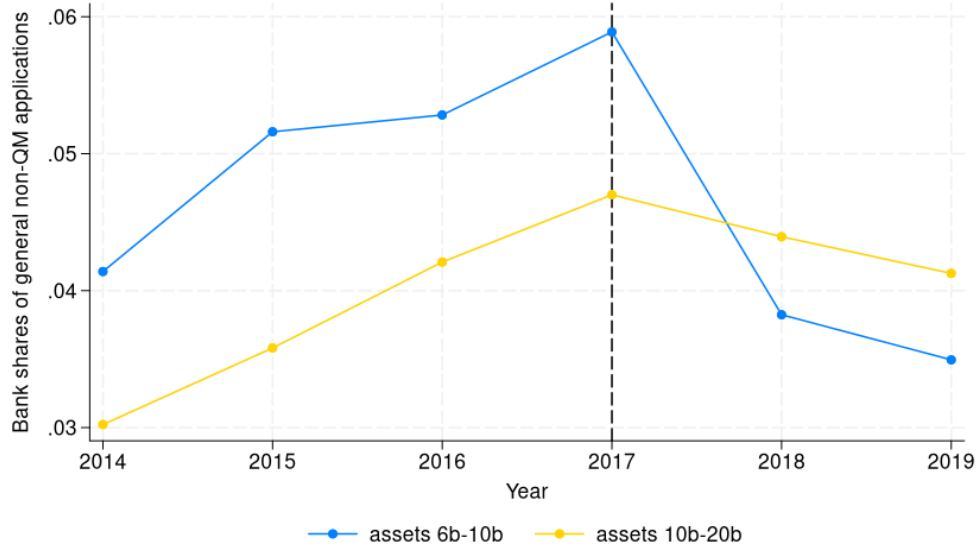


Note: The figure plots the observed means of the natural log of the number of generally non-QM applications received by banks each year, averaged across banks above and below \$10 billion in assets in 2017. The figure is based on the bank-year sample which comes from aggregating the the full sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications at banks with assets between \$6 and \$20 billion in 2017. General non-QM refers to loans that are above the conforming loan limit and have LTI in the top quartile among all HMDA/LAR records over the sample period.

5.3.2 Regression Results

Table 5 reports regression results for equation 4 estimated using the bank-year sample. Columns (1) and (2) report results for the outcome $\ln(\text{Number of General Non-QM Applications})$ which are not statistically significant. Columns (3) and (4) report results for the share of applications that are general non-QM, finding a statistically significant negative coefficient on $\mathbb{1}\{Assets < \$10b\} \times Post$.

Figure 4: Bank Level Shares of General Non-QM Applications Trends



Note: The figure plots the observed means of the share of applications that are for generally non-QM loans received by banks each year, averaged across banks above and below \$10 billion in assets in 2017. The figure is based on the bank-year sample which comes from aggregating the the full sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications at banks with assets between \$6 and \$20 billion in 2017. General non-QM refers to loans that are above the conforming loan limit and have LTI in the top quartile among all HMDA/LAR records over the sample period.

This suggests that the reduction in CFPB enforcement actions had a greater effect on the share of general non-QM applications received than the creation of the EGRRCPA QM. However, the results become statistically insignificant when the banks are propensity score reweighted to improve comparability (table C3).

6 Interpretation

Overall, this paper finds evidence that the relaxation of the CFPB's regulatory philosophy had a larger effect than the creation of the EGRRCPA QM on the supply of general non-QM loans, measured as approval rates and bank shares of applications. It is surprising that the EGRRCPA QM, a policy targeted specifically at banks' general non-QM lending, had a smaller effect than the general reduction of CFPB oversight, which was not aimed at non-QM loans.

One explanation for this effect is that supervised lenders pay more attention to the rules their regulator enforces and may, therefore, be more likely to adapt to changes in the enforcement of those rules than unsupervised lenders. Banks supervised by the CFPB may have been more aware

Table 5: Bank Level Regression Results

	ln(Gen Non-QM Apps)		Gen Non-QM App Share	
	(1)	(2)	(3)	(4)
$\mathbb{1}\{Assets < \$10b\} \times Post$	-0.011 (0.179)	-0.055 (0.181)	-0.010** (0.004)	-0.011** (0.004)
Constant	3.634*** (0.039)	4.014*** (0.690)	0.046*** (0.001)	0.041** (0.016)
Year FE	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes
Covariates Included [†]		Yes		Yes
Observations	610	610	610	610
R-squared	0.877	0.888	0.867	0.900

Notes: [†]The covariates included are bank-level averages for 2014 - 2017 of applicant income, loan amounts, share of joint applications, share of refinancing applications, share of manufactured homes, share of Black applicants, and share of Hispanic applicants. Robust standard errors clustered at the bank level in parentheses, *** p<0.01, ** p<0.05, * p<0.1. Results for equation 4 estimated using the bank-year sample are reported. The bank-year sample aggregates the full sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications from 2014 to 2019 at lenders with assets between \$6 and \$20 billion in 2017 to the bank-year level. General non-QM refers to loans that are non-conforming (above the conforming loan limit) and have LTI in the top quartile among all HMDA/LAR records over the sample period. Columns (1)-(2) report results for outcome as the natural log of the number of generally non-QM loan applications received by banks each year, columns (3)-(4) report results for the bank-level annual shares of applications that are generally non-QM, and columns (5)-(6) report results for the bank-level annual shares of approved loans that are generally non-QM.

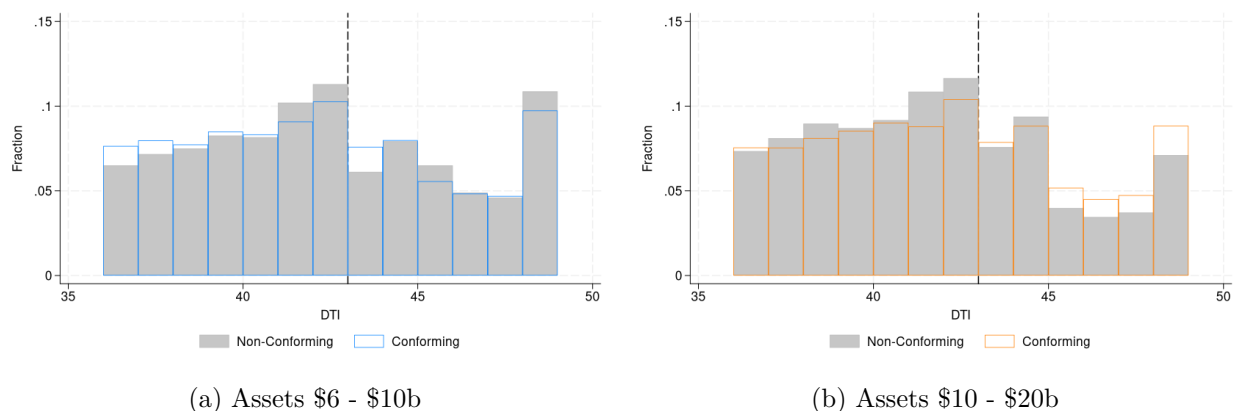
that the ATR/QM rule is enforced by the CFPB and also have noted the change in CFPB leadership meant that this rule was unlikely to be enforced. Banks that are not supervised by the CFPB may have added the QM rule to their loan approval process when the rule was first implemented in response to the CFPB's wide reaching information campaign about the rule. However, these banks may have lost track of where the rule came from and may not have been aware that the EGRRCPA QM meant that this rule no longer applied to them.

Figure 5 provides suggestive evidence for this mechanism. In 2022, the general definition of a QM changed from loans that are conforming or have a DTI below 43 percent to loans with an annual percent rate (APR) below the average prime offer rate (APOR) for similar loans plus 2.25 percentage points.²⁰ Panel 5a plots the DTI distribution of approved conforming and non-conforming loans at banks with assets \$6 - \$10 billion in 2023, while panel 5b plots the 2023 DTI distribution for banks with assets \$10 - \$20 billion. At smaller banks, non-conforming loans display greater bunching at the 43 percent DTI threshold relative to conforming loans. The QM definition that called for non-conforming loans to have a DTI below 43 percent has not been binding for small banks since

²⁰Appendix A contains more information about the QM categories over time.

the 2018 creation of the EGRRCPA QM. However, five years later, these banks still display more bunching at this 43 percent DTI threshold than the CFPB regulated banks for which the 43 percent DTI threshold was removed from the QM definition only one year earlier. This suggests that banks not regulated by the CFPB may have greater reliance on “rules of thumb” when making lending decisions and that these rules of thumb may be more persistent for these banks.

Figure 5: Bunching at the 43 percent DTI Threshold by Bank Size in 2023



Note: This figure plots the distribution of DTI for conforming and non-conforming approved, retained loans in 2023 at banks with assets between \$6 and \$10 billion and banks with assets between \$10 and \$20 billion. The sample is limited to conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications.

7 Conclusion

This paper studies the net effect on mortgage lending of the concurrent introductions of the EGRRCPA QM and the reduction in CFPB enforcement activity. Using HMDA loan-level data and a difference-in-differences style framework comparing banks just above and below the \$10 billion asset threshold, I find that the decline in CFPB oversight had a greater effect on general non-QM mortgage lending than the EGRRCPA QM.

The analysis provide suggestive evidence in support of an important mechanism: supervised institutions appear more responsive to changes in the enforcement of rules than unsupervised institutions are to changes in rules. The 2022 revision to the QM definition shows greater persistence in “rules-of-thumb” underwriting among non-supervised banks, while supervised banks adjusted more sharply once regulatory guidance changed. This asymmetry suggests that lenders do not necessarily adjust their underwriting in response to all regulatory changes, especially when prior

guidelines remain embedded in their processes.

Taken together, the findings imply that supervisory enforcement can exert stronger influence on mortgage credit supply than rule changes, even when those rule changes are directly targeted at a specific market segment. For policymakers, this suggests that the effectiveness of deregulation depends not only on formal rule design but also on how institutions internalize the reforms.

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A Appendix: Qualified Mortgage Classifications

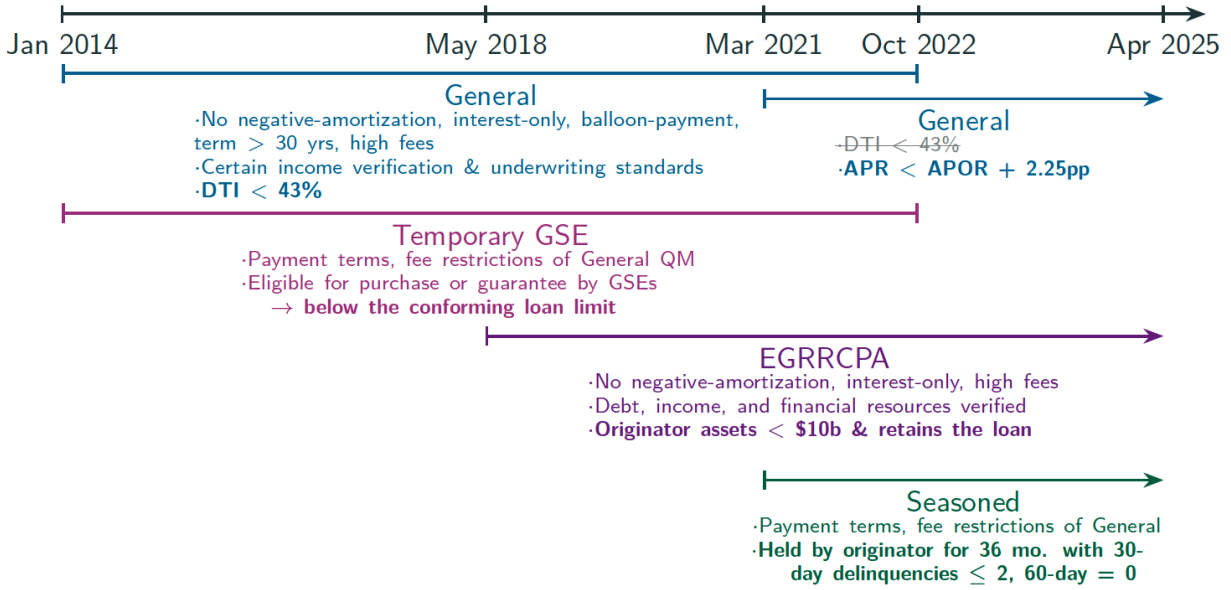
Figure A1 provides an overview of QM categories and their evolving requirements. The ATR/QM rule went into effect in January 2014 with two main types of QMs: General and Temporary Government Sponsored Enterprise (GSE).²¹ Both prohibited negative-amortization, interest-only, and balloon-payment features, terms that exceed 30 years, and excessive points and fees. General QMs imposed certain income verification and underwriting standards as well as requiring that the Debt-to-Income (DTI) ratio of the loan not exceed 43 percent. The Temporary GSE category included all loans that were eligible for purchase or guarantee by the GSEs, Fannie Mae and Freddie Mac. While the GSEs impose certain income verification and underwriting standards, the primary criterion in practice was that the loan amount was below the conforming loan limit. Since the vast majority of mortgages are eligible for purchase by the GSEs, the GSE QM category was intended as a temporary patch to ensure the smooth implementation of the ATR/QM rule.

There were several later changes to the QM categories. As the expiration date for the Temporary GSE QM category approached, regulators were concerned about their continued prevalence. In response, the CFPB revised the standards for General QMs and created a new category called Seasoned QMs in 2021. The main change to General QMs was replacing the DTI threshold with a requirement that the APR not exceed the Average Prime Offer Rate (APOR) by more than 2.25 percentage points. To the extent that price captures the riskiness of the loan, being within a certain threshold of the APOR should signal sufficiently low risk. This change took effect on March 1st, 2021, but the mandatory compliance date was not until October 1st, 2022, meaning both definitions were in use during that 19 month period.²² Seasoned QMs provide a pathway for non-QM loans to attain QM status if they remain on the originator’s balance sheet for 36 months with no more than two 30-day delinquencies and zero 60-day delinquencies. During the “seasoning period,” borrowers demonstrate that they are able to repay the debt. The effects of these later changes to the QM categories have not been formally studied.

²¹Mortgages issued by the Federal Housing Administration (FHA), U.S. Department of Agriculture (USDA), and Department of Veterans Affairs (VA) are also considered QMs. Additionally, loans with certain payment and fees features issued by lenders with total assets below \$2 billion that originated fewer than 2,000 first-lien, closed-end residential mortgages are considered Small Creditor QMs. These loans will be excluded from the analysis of this paper.

²²There is some evidence that market participants were hesitant to switch from the DTI threshold to the price-based definition for General QMs (Adelson, 2021).

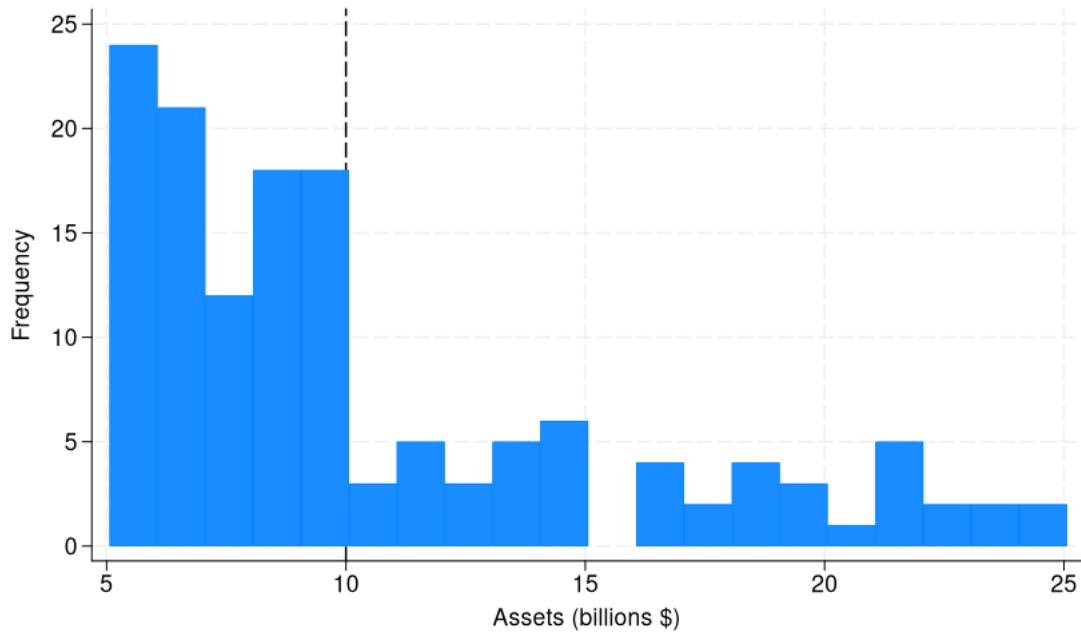
Figure A1: Qualified Mortgage (QM) requirements by type over time.



Note: Mortgages issued by the Federal Housing Administration (FHA), U.S. Department of Agriculture (USDA), and Department of Veterans Affairs (VA) are also considered QMs throughout this period, but are not included in this diagram. Small Creditor QMs issued by lenders with total assets below \$2 billion that originated fewer than 2,000 first-lien, closed-end residential mortgages are also omitted.

B Appendix: Asset Distribution

Figure B1: Distribution of Assets in 2017 for Depository Institutions, Assets \$5 - \$25 billion



Source: HMDA Lender File

C Appendix: Propensity Score Reweighting

To improve comparability between treated and control banks and ensure more credible pre-trend similarities, I implement a propensity-score reweighting procedure following standard approaches in the literature. While banks above and below the \$10 billion asset threshold necessarily differ in size, they may also systematically differ in other characteristics, such as geography, portfolio composition, and pre-reform trends. I reweight the banks above the \$10 billion asset threshold to emphasize the banks that look more like those below the \$10 billion threshold by implementing the following procedure:

1. Construct a bank-level dataset by aggregating the loan-level data. The dataset contains an indicator for whether the bank is above or below the \$10 billion threshold and the pre-period (2014 - 2017) averages for: loan approval, loan retention, share of general non-QM applications, share of general non-QM loans, share of loan applications from each census region (Northeast, Midwest, South, West), applicant incomes, loan amounts, share of joint applications, share of refinancing applications, share of manufactured home applications, share of Black applicants, and share of Hispanic applicants. The dataset also contains the pre-period linear trend for each of these variables. Denote this vector of preperiod characteristics for each bank b as X_b .
2. In the pooled sample of treated and untreated banks, I estimate a logit model of the probability that the bank has assets below \$10 billion in 2017 using the characteristics listed in the bullet point above:

$$\Pr(\mathbb{1}\{Assets_{b,2017} < \$10b\} = 1) = p(X_b)$$

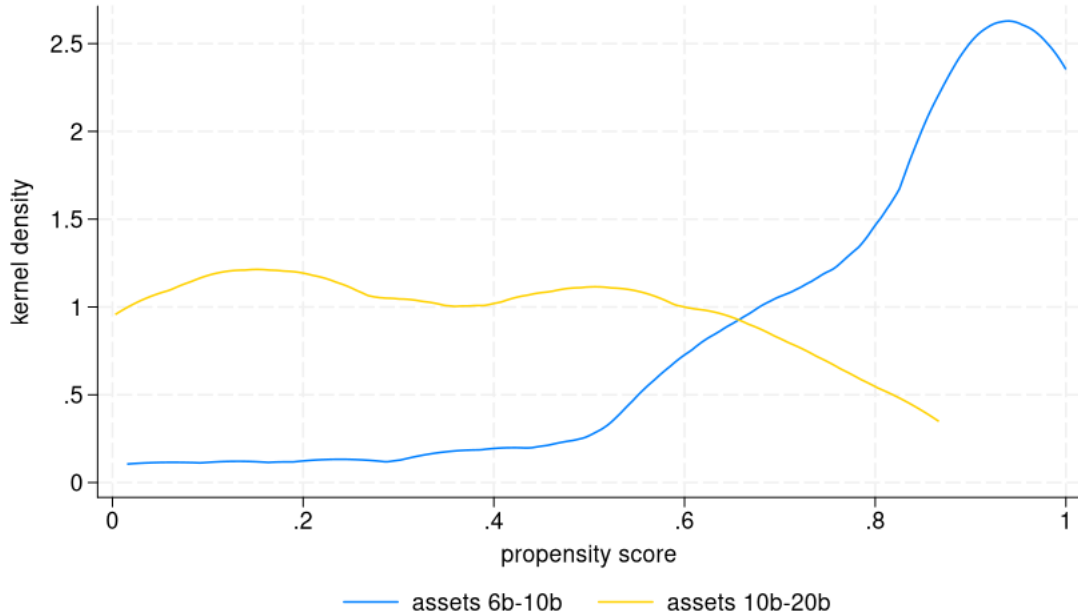
The predicted probability of treatment for each bank is then generated from the logit. This fitted value $\hat{p}(X_b)$ is the propensity score.

3. I then construct the weights as

$$w_b = \begin{cases} 1 & \text{if } \mathbb{1}\{Assets_{b,2017} < \$10b\} = 1 \\ \frac{\hat{p}(X_b)}{1 - \hat{p}(X_b)} & \text{if } \mathbb{1}\{Assets_{b,2017} < \$10b\} = 0 \end{cases}$$

4. The sample is then trimmed for common support. Figure C1 plots the distribution of propensity scores for banks above and below the \$10 billion asset threshold. To avoid extrapolation and ensure overlap, I restrict the sample to banks with estimated propensity scores in the region where both groups' score distributions overlap. This trimming removes banks with extreme propensity scores that lack comparable counterparts in the opposite group.
5. I then merge these weights back into the loan and bank-year level datasets. In the loan-level regressions, each loan inherits its bank's weight.

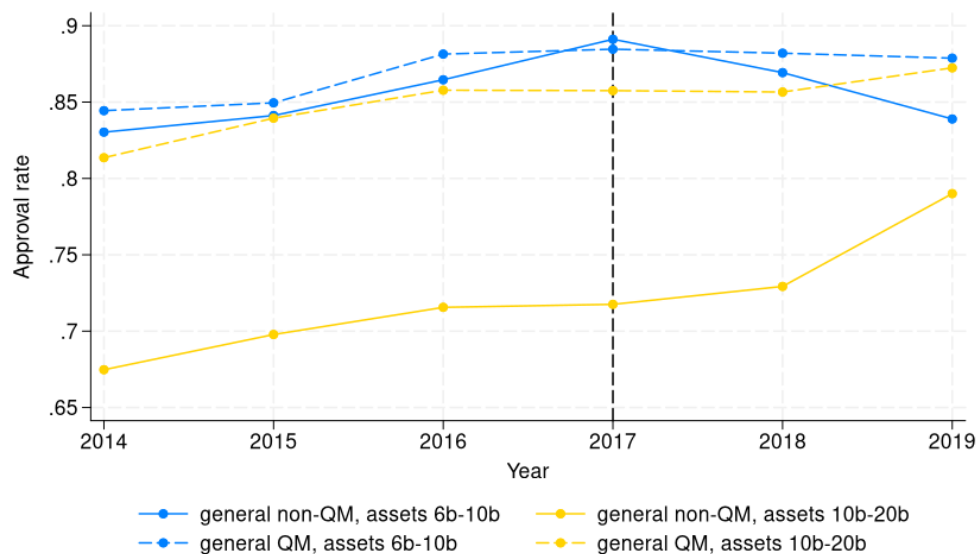
Figure C1: Propensity Score Kernel Density by Bank Size



To assess how these weights affect the pre-trends, I plot the weighted means over time by bank size for the loan level outcomes: approval rates (figure C2) and loan retention (figure C3), and the bank-level outcomes: $\ln(\text{general non-QM applications})$ (figure C4) and general non-QM application share (figure C5). This propensity score reweighting generally improves pre-trends compared to the

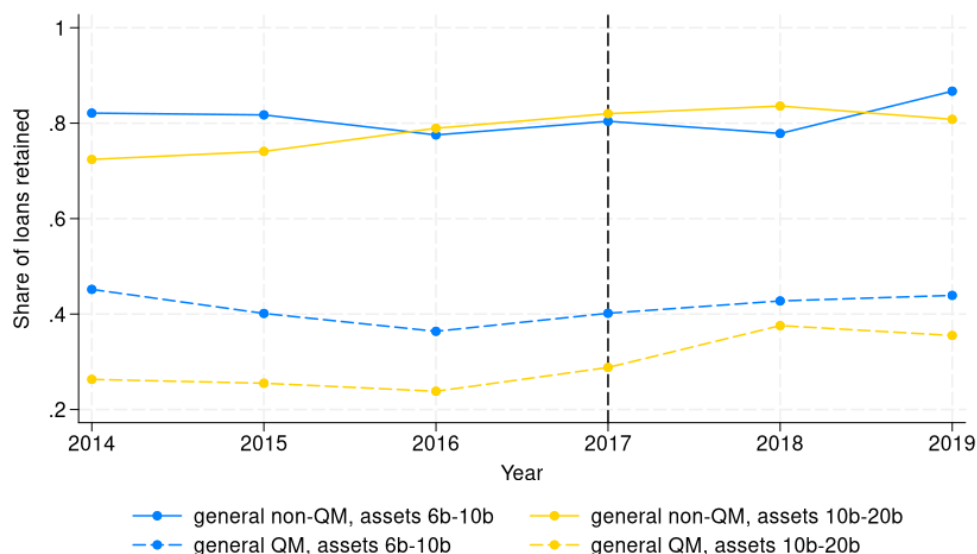
analogous plots in section 5 of the main text of the paper. However, this reweighted analysis has fewer observations as the result of trimming for common support. Re-weighted regression results are reported in table C1 for approval rates, table C2 for loan retention, and table C3 for bank level regressions.

Figure C2: Mortgage Approval Rate Trends by Bank Size and Market Segment, Propensity Score Re-weighted



Note: The figure plots the weighted means of approval rate by bank size (treatment status) and market segment (general non-QM and general QM) for the full sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications at banks with assets between \$6 and \$20 billion in 2017. General non-QM refers to loans that are above the conforming loan limit and have LTI in the top quartile among all HMDA/LAR records over the sample period.

Figure C3: Mortgage Retention Rate Trends by Bank Size and Market Segment, Propensity Score Re-weighted



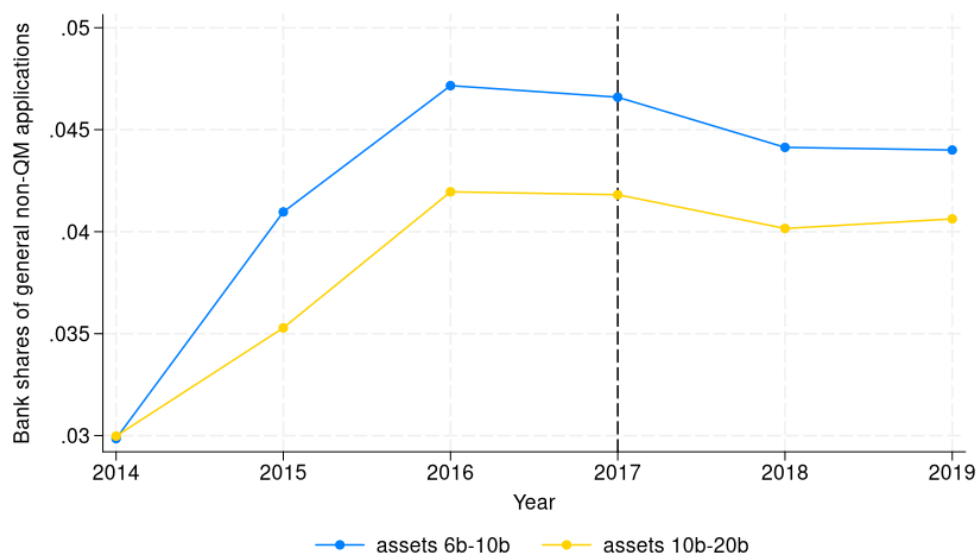
Note: The figure plots the weighted means of loan retention rate by bank size (treatment status) and market segment (general non-QM and general QM) for the full sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications at banks with assets between \$6 and \$20 billion in 2017. General non-QM refers to loans that are above the conforming loan limit and have LTI in the top quartile among all HMDA/LAR records over the sample period. Loans are considered retained if the bank does not sell them within the calendar year of origination.

Figure C4: Bank Level $\ln(\text{General Non-QM Applications})$ Trends, Propensity Score Re-weighted



Note: The figure plots the weighted means of the natural log of the number of generally non-QM applications received by banks each year, averaged across banks above and below \$10 billion in assets in 2017. The figure is based on the bank-year sample which comes from aggregating the the full sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications at banks with assets between \$6 and \$20 billion in 2017. General non-QM refers to loans that are above the conforming loan limit and have LTI in the top quartile among all HMDA/LAR records over the sample period.

Figure C5: Bank Level Shares of General Non-QM Applications Trends, Propensity Score Re-weighted



Note: The figure plots the weighted means of the share of applications that are for generally non-QM loans received by banks each year, averaged across banks above and below \$10 billion in assets in 2017. The figure is based on the bank-year sample which comes from aggregating the the full sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications at banks with assets between \$6 and \$20 billion in 2017. General non-QM refers to loans that are above the conforming loan limit and have LTI in the top quartile among all HMDA/LAR records over the sample period.

Table C1: Mortgage Approval Rate Propensity Score Re-weighted Regressions

	Diff-in-Diff Style			Triple Diff Style		
	(1)	(2)	(3)	(4)	(5)	(6)
$1 \{Assets < \$10b\} \times Post \times GenNonQM$				-0.048*	-0.044*	-0.045**
				(0.024)	(0.022)	(0.022)
$1 \{Assets < \$10b\} \times Post$	-0.021	-0.016	-0.012	0.009	0.010	0.012
	(0.025)	(0.021)	(0.018)	(0.012)	(0.011)	(0.010)
$Post \times GenNonQM$				0.035**	0.030**	0.028**
				(0.015)	(0.014)	(0.013)
$1 \{Assets < \$10b\} \times GenNonQM$				0.082**	0.081**	0.084**
				(0.039)	(0.036)	(0.037)
$GenNonQM$				-0.133***	-0.125***	-0.126***
				(0.035)	(0.020)	(0.021)
Joint Application		0.009	0.006		0.008*	0.007
		(0.008)	(0.008)		(0.005)	(0.004)
Refinancing		-0.116***	-0.114***		-0.061***	-0.061***
		(0.008)	(0.008)		(0.012)	(0.011)
Manufactured Home		-0.362***	-0.392***		-0.140***	-0.151***
		(0.051)	(0.048)		(0.014)	(0.018)
Black		-0.092***	-0.100***		-0.075***	-0.073***
		(0.017)	(0.015)		(0.006)	(0.006)
Hispanic		-0.014	-0.015		-0.039***	-0.038***
		(0.018)	(0.019)		(0.006)	(0.006)
Constant	0.771***	0.819***	0.824***	0.855***	0.887***	0.893***
	(0.002)	(0.005)	(0.006)	(0.001)	(0.008)	(0.007)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE		Yes	Yes		Yes	Yes
Income Centile		Yes	Yes		Yes	Yes
Amount Centile		Yes	Yes		Yes	Yes
2014 Omitted			Yes			Yes
Observations	44,907	44,601	39,935	1,560,194	1,560,149	1,336,468
R-squared	0.074	0.222	0.228	0.033	0.080	0.081

Note: Robust standard errors clustered at the bank level in parentheses, *** p<0.01, ** p<0.05, * p<0.1. Columns (1)-(3) report results for equation 1 estimated using general non-QM loan applications in the sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications from 2014 to 2019 at lenders with assets between \$6 and \$20 billion in 2017. General non-QM refers to loans that are non-conforming (above the conforming loan limit) and have LTI in the top quartile among all HMDA/LAR records over the sample period. Columns (4)-(6) report results for equation 3 with loan approval as the outcome, estimated using the full sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications from 2014 to 2019 at lenders with assets between \$6 and \$20 billion in 2017. Applications have been propensity score reweighted by bank as described in the text of appendix C.

Table C2: Mortgage Retention Rate Propensity Score Re-weighted Regressions

	Diff-in-Diff			Triple Diff		
	(1)	(2)	(3)	(4)	(5)	(6)
$1 \{Assets < \$10b\} \times Post \times GenNonQM$				0.016 (0.056)	-0.000 (0.051)	0.014 (0.054)
$1 \{Assets < \$10b\} \times Post$	-0.011 (0.039)	-0.028 (0.035)	-0.007 (0.036)	-0.053** (0.022)	-0.048** (0.022)	-0.043* (0.022)
$Post \times GenNonQM$				-0.029 (0.043)	-0.026 (0.041)	-0.037 (0.042)
$1 \{Assets < \$10b\} \times GenNonQM$				-0.020 (0.075)	-0.013 (0.062)	-0.027 (0.065)
$GenNonQM$				0.382*** (0.063)	0.146*** (0.044)	0.157*** (0.046)
Joint Application		-0.012* (0.006)	-0.014* (0.007)		-0.009** (0.004)	-0.009** (0.004)
Refinancing		0.008 (0.029)	0.008 (0.030)		0.023** (0.010)	0.017 (0.011)
Manufactured Home		0.082** (0.035)	0.111** (0.049)		0.176*** (0.047)	0.176*** (0.047)
Black		0.009 (0.015)	0.010 (0.016)		0.065** (0.025)	0.067*** (0.025)
Hispanic		-0.017 (0.012)	-0.012 (0.011)		0.014 (0.010)	0.016 (0.011)
Constant	0.800*** (0.004)	0.806*** (0.012)	0.808*** (0.012)	0.332*** (0.002)	0.327*** (0.006)	0.332*** (0.006)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE		Yes	Yes		Yes	Yes
Income Centile		Yes	Yes		Yes	Yes
Amount Centile		Yes	Yes		Yes	Yes
2014 Omitted			Yes			Yes
Observations	34,846	34,570	31,123	1,310,777	1,310,669	1,131,691
R-squared	0.251	0.314	0.313	0.260	0.330	0.319

Note: Robust standard errors clustered at the bank level in parentheses, *** p<0.01, ** p<0.05, * p<0.1. Columns (1)-(3) report results for equation 2 estimated using general non-QM approved loans in the sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications from 2014 to 2019 at lenders with assets between \$6 and \$20 billion in 2017. General non-QM refers to loans that are non-conforming (above the conforming loan limit) and have LTI in the top quartile among all HMDA/LAR records over the sample period. Columns (4)-(6) report results for equation 3 with loan retention as the outcome, estimated using the full sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications from 2014 to 2019 at lenders with assets between \$6 and \$20 billion in 2017. Loans are considered retained if the bank does not sell them within the calendar year of origination. Loans have been propensity score reweighted by bank as described in the text of appendix C.

Table C3: Bank Level Propensity Score Re-weighted Regression Results

	ln(Gen Non-QM Apps)		Gen Non-QM App Share	
	(1)	(2)	(3)	(4)
$\mathbb{1}\{Assets < \$10b\} \times Post$	0.116 (0.262)	0.152 (0.232)	-0.003 (0.004)	-0.002 (0.004)
Constant	4.011*** (0.037)	4.356*** (0.530)	0.040*** (0.001)	0.034** (0.013)
Year FE	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes
Covariates Included [†]		Yes		Yes
Observations	343	343	343	343
R-squared	0.861	0.891	0.890	0.912

Notes: [†]The covariates included are bank-level averages for 2014 - 2017 of applicant income, loan amounts, share of joint applications, share of refinancing applications, share of manufactured homes, share of Black applicants, and share of Hispanic applicants. Robust standard errors clustered at the bank level in parentheses, *** p<0.01, ** p<0.05, * p<0.1. Results for equation 4 estimated using the bank-year sample are reported. The bank-year sample aggregates the full sample of conventional, complete/not-withdrawn, first-lien home purchase and refinancing mortgage applications from 2014 to 2019 at lenders with assets between \$6 and \$20 billion in 2017 to the bank-year level. General non-QM refers to loans that are non-conforming (above the conforming loan limit) and have LTI in the top quartile among all HMDA/LAR records over the sample period. Columns (1)-(2) report results for outcome as the natural log of the number of generally non-QM loan applications received by banks each year, columns (3)-(4) report results for the bank-level annual shares of applications that are generally non-QM, and columns (5)-(6) report results for the bank-level annual shares of approved loans that are generally non-QM. Loans have been propensity score reweighted by bank as described in the text of appendix C.